

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

4.2 Fact 2: Though overall employment continues to grow, employment growth for young workers in particular has been stagnant

Figure 4 shows overall employment trends across age groups, pooling together all occupations. Overall employment remains robust, coinciding with a low national unemployment rate in the post-pandemic period. However, Figure 4 suggests some leveling off in employment growth for young workers relative to other age groups, consistent with recent discussion of a potentially worsening job market for entry-level workers (Chen, 2025; Federal Reserve Bank of New York, 2025).

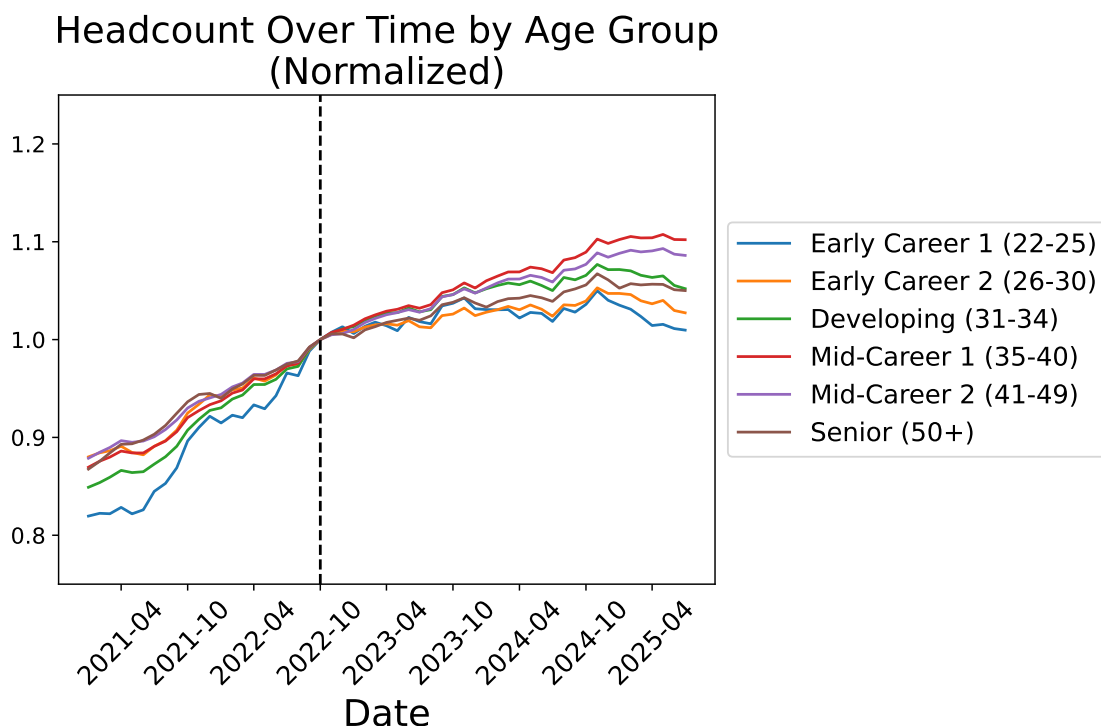


Figure 4: Employment changes by age. Including all occupations.

Figure 5 offers insight into how these trends relate to AI exposure. For each age group, employment growth from late 2022 to July 2025 was 6-13% for the lowest three AI exposure quintiles, with no clear ordering in employment growth by age. In contrast, for the highest two exposure quintiles employment for 22-25 year olds declined by 6% between late 2022 and July 2025, while employment for workers aged 35-49 grew by over 9%. These results show that declining employment in AI-exposed jobs is driving tepid overall employment growth for workers between the ages of 22 and 25.

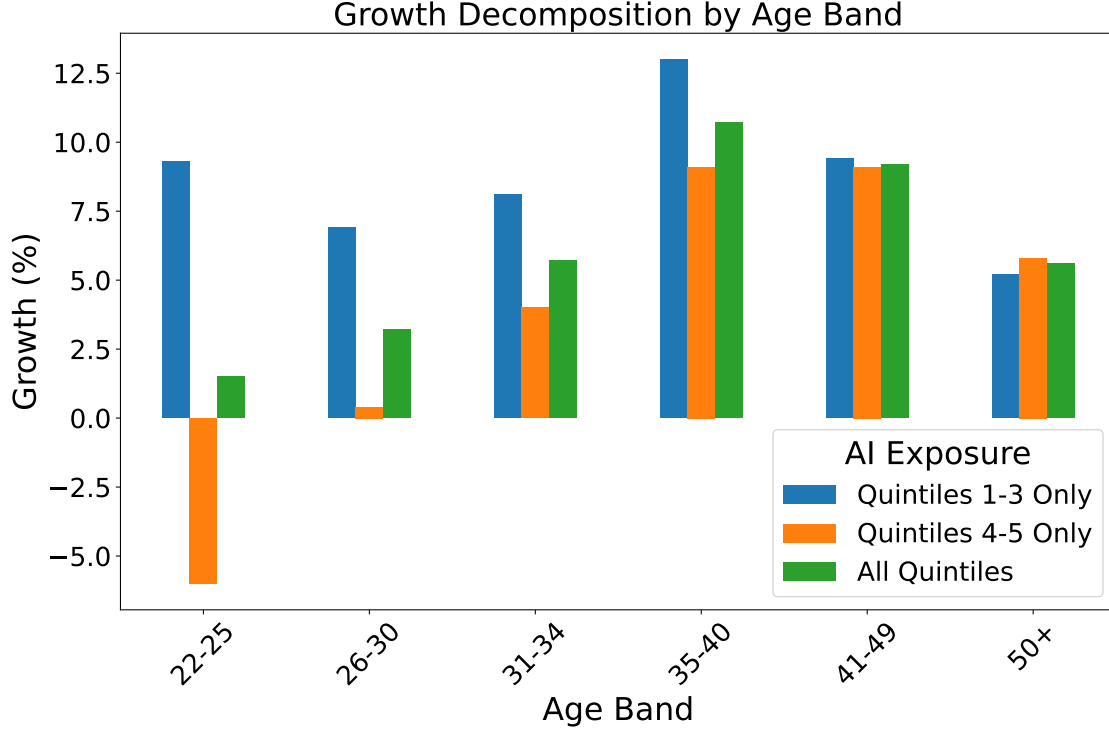


Figure 5: Growth in employment between October 2022 and July 2025 by age and GPT-4 β -based AI exposure group.

While these findings suggest divergent employment outcomes by AI exposure for young workers, we caution the trends observed in these first two facts could be driven by other changes in the US economy. Our subsequent facts evaluate the robustness of the results to alternative analyses.

4.3 Fact 3: Entry-level employment has declined in applications of AI that *automate* work, with muted changes for *augmentation*

AI exposure can either complement or substitute for labor. These may have very different implications for the labor market (Brynjolfsson, 2022).

To assess how employment patterns differ based on the complementarity or substitutability of AI with labor, we use data on generative AI usage from the Anthropic Economic Index (Handa et al., 2025). The Index provides an estimate of the share of queries that pertain to each occupation. In addition, for each task it reports estimates of the share of queries pertaining to that task that are “automative,” “augmentative,” or none of the above. We use this information as an estimate of

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁷

We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁸ Standard errors are clustered by firm.

Results are in Figure 9, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 12 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 3. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁷Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

¹⁸Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.